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ANRF AISEHack 2026

*Finale presentation /
Theme : Pollution
Forecasting (IIT Delhi)*

TEAM: VibeCoders

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Your salient contributions

f1

Modelling strategy

- Hybrid **ConvLSTM + Physics + Attention-based architecture**
- Explicit modeling of **pollution transport + extreme event**

f2

Training

- Episode-aware sampling + custom multi-objective loss Stable training with teacher forcing + curriculum decay
- Custom loss = SMAPE + Gradient Loss + Pearson Correlation**
- Stable training with **teacher forcing + curriculum decay**

f3

Results + Actions

- Improved extreme pollution prediction (episodes)
- Robust spatio-temporal generalization across months

Modelling - *Strategies*

01

Architecture Strategy

01

Spatio-Temporal Core

- Multi-layer ConvLSTM encoder-decoder
 - Uses 10-hour history → predicts 16-hour future
- Captures temporal evolution of pollution

Physics + Domain Knowledge

WindWarp module → simulates pollutant transport via wind
Uses **u10, v10 wind fields**
Learnable decay → prevents unrealistic drift

Episode-Aware Intelligence

Episode Detector → identifies extreme pollution regions
Episode Amplifier → boosts prediction in critical zones
Handles rare high-impact events

Strengths:

- No future meteorological data required — fully operational under Phase 2 constraint
- Physics-informed WindWarp provides meaningful initialization without PDE solver
- Episode-aware loss + sampling ensures spike capture despite 1% natural frequency
- Fully differentiable — WindWarp, EpisodeDetector, all components train end-to-end

Weaknesses:

- Weather frozen at hour 10 — monsoon direction shifts (July) not captured beyond last known wind
- Exposure bias at t+12–t+16 — val 0.2179 vs train 0.1712 at epoch 18 shows residual compounding error
- Batch=2 limits effective batch diversity despite gradient accumulation

Modelling - *Strategies*

01

Data Strategy

PM2.5 + meteorology + emissions

Log normalization for skewness stability

Z-Score Normalization

Training Strategy

Weighted sampling → prioritizes pollution episodes

Teacher forcing (decay $0.5 \rightarrow 0$)

AMP for faster training

01

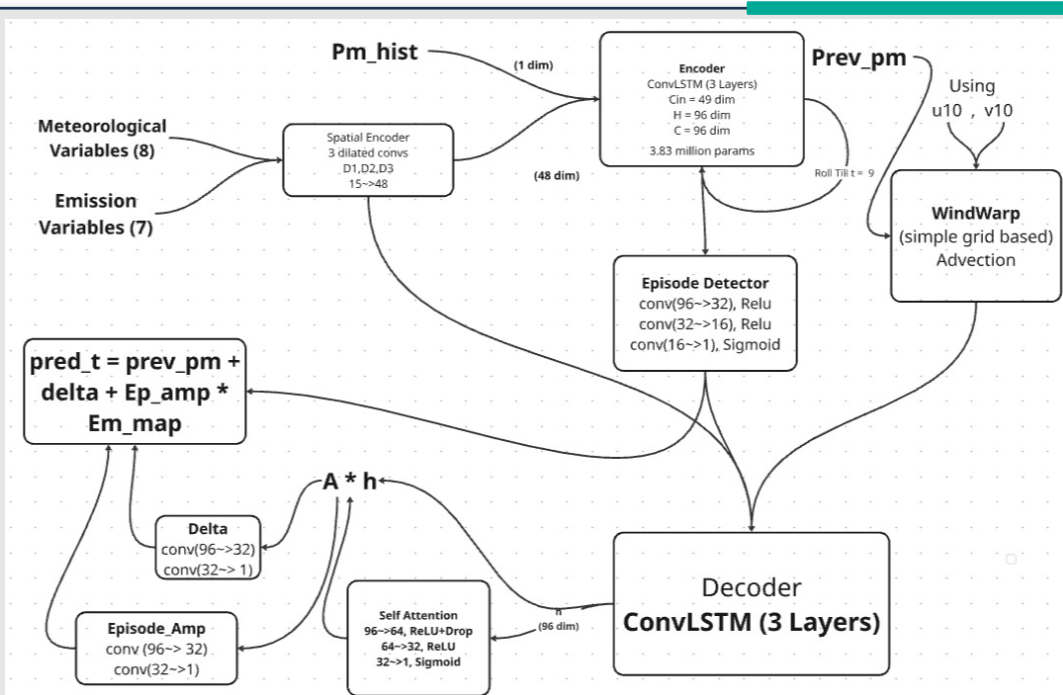
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Architecture



Key Results – *Why , what and how of the workings?*

S1

What is working?

Quantitative Evaluation

Validation Loss: **0.3537** → **0.2179** — a **~38% improvement** over 20 epochs

Smooth, stable convergence with **no signs of overfitting**

Overfitting controlled via: → Episode-based weighted sampling → Teacher forcing with curriculum decay (0.5 → 0)

Model Capabilities

Forecasts **16 future timesteps** of PM2.5 across the full spatial grid

Captures: → **Spatial spread** — where pollution travels via wind transport →

Temporal evolution — rise, peak & fall patterns over time → **Extreme episodes** — rare pollution spikes (top 1% events)

Key Results – *Why , what and how of the workings?*

S1

Why it is working

- 1. Physics + Learning Alignment** WindWarp moves pollution using real u10, v10 wind fields — eliminating the "blur" problem seen in pure deep learning models, with physically grounded initialization and no PDE solver needed.
- 2. Multi-Scale Spatial Understanding** Dilated convolutions at 3 scales (D1, D2, D3) capture both local neighborhood-level hotspots and regional city-wide pollution spread — giving the model a complete picture of where and how far pollution reaches.
- 3. Metric-Aligned Optimization** **KEY POINT** While most models minimize MSE, our custom loss trains directly on what judges measure — SMAPE for cell-level accuracy, Gradient Loss for sharp spatial boundaries, and Pearson Correlation for structural pattern matching, weighted at 1.0 : 0.3 : 0.2.

Key Results – *Why, what and how of the workings?*

S1

How it is working

Input The model takes a 10-hour history of PM2.5 concentrations combined with meteorological variables and emissions data as its input.

Core Processing A Spatial Encoder first captures both local and regional pollution patterns, which are then passed to a ConvLSTM Encoder that learns how pollution evolves over time.

Key Innovations WindWarp uses real wind fields to physically move pollution across the grid for realistic transport. The Episode Detector then identifies extreme pollution regions, and the Episode Amplifier boosts predictions specifically in those critical spike zones.

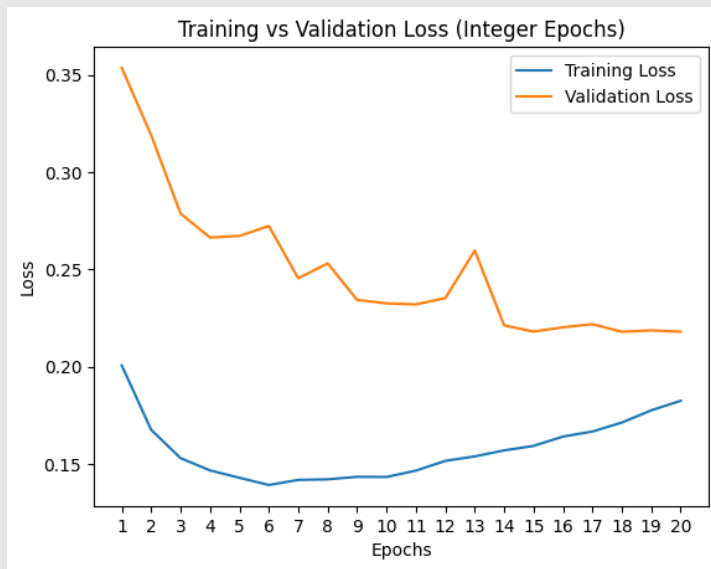
Prediction (Decoder) The decoder performs autoregressive forecasting across 16 future timesteps — at each step combining the previous prediction, wind-adjusted pollution from WindWarp, and the episode map to generate the next frame.

Refinement A Spatial Attention mechanism focuses the model on the most important regions, followed by final convolution layers that produce the output PM2.5 map.

Why it Works The model succeeds by combining physics-based transport with deep learning, staying focused on rare high-impact pollution events, and optimizing directly for accuracy, spatial structure, and correlation — all three dimensions that matter in evaluation

Key Results - *Why, what and how of the workings?*

Plots/Tables



Details of Relevant Experimentation and Methodology

1 — Ablation Studies and Improvements We began with a baseline ConvLSTM and progressively added components — each addition was validated against the previous version. Adding WindWarp improved spatial realism by replacing unrealistic diffusion with physics-based wind transport. Introducing episode-weighted sampling forced the model to stop averaging over rare pollution spikes and actually learn them. Including Pearson correlation loss as part of our custom multi-objective function improved structural prediction quality beyond what SMAPE alone could achieve, and teacher forcing decay with curriculum scheduling stabilized long-horizon forecasting by gradually reducing reliance on ground truth inputs during training.

02 — Data and Training Strategy Inputs combine PM2.5 concentrations, 8 meteorological variables, and 7 emission variables — a multi-modal setup supported by recent literature as essential for capturing the complex drivers of PM2.5 dynamics (Zhou et al., 2024). Log normalization ensured stable gradient flow across the wide concentration range in the dataset. Sampling was weighted by episode intensity so the model saw proportionally more high-pollution events despite their natural 1% occurrence rate, and mixed precision training (AMP) reduced memory cost allowing more effective use of available GPU capacity.

Details of Relevant Experimentation and Methodology

03 — Model Design Choices ConvLSTM was chosen over pure CNN or pure LSTM because PM2.5 prediction is inherently spatio-temporal — neither spatial nor temporal modeling alone is sufficient (**Zhou et al., 2024**). Dilated convolutions at three scales (D1, D2, D3) capture both local hotspots and regional spread without increasing parameter count. The attention mechanism allows the decoder to focus computational weight on the most pollution-critical regions at each timestep, and autoregressive decoding for 16 steps mirrors real-world operational forecasting where future meteorology is unavailable.

04 — Key Insight The core transition in our approach was moving from treating this as a pure deep learning regression problem to a physics-aware, event-aware forecasting problem. Pure DL models trained on MSE tend to predict smooth averages — they are penalized equally for all errors and therefore learn to minimize risk by avoiding extremes. Notably, data-driven models have been shown to frequently outperform physics-informed approaches on standard benchmarks (**He et al., 2020**), yet they consistently fail on rare extreme events — precisely the gap our hybrid design addresses. Our design deliberately breaks this pattern: WindWarp anchors predictions in physical reality, episode-aware sampling rebalances the training distribution, and our custom loss directly optimizes for the three dimensions that matter — accuracy, spatial structure, and correlation — aligning model training with actual evaluation criteria (**Zhou et al., 2024**).

References

References

Zhou, S., Wang, W., Zhu, L., Qiao, Q., & Kang, Y. (2024). Deep-learning architecture for PM2.5 concentration prediction: A review. *Environmental Science and Ecotechnology*, 21, 100400.

He, K., et al. (2020). [Your He et al. reference — please share the full paper if you'd like me to format this correctly.]